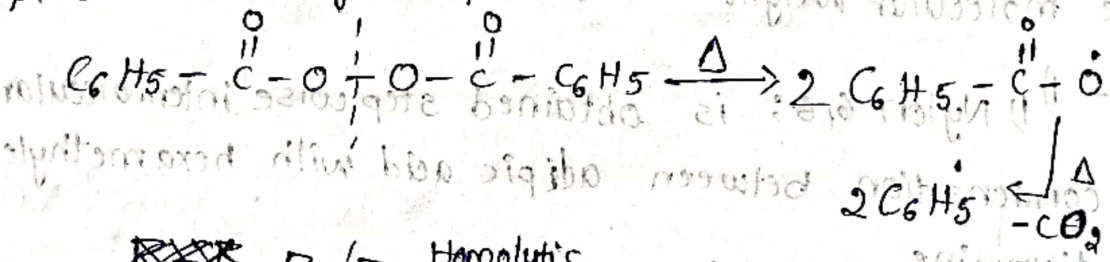
Mechanism of free radical addition polymerisation:

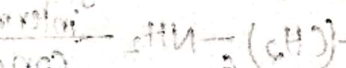
Mechanism of free radical addition polymerisation involves 3 steps

1) Initiation 2) Propagation 3) Terminator.

1) **Initiation**:- The initiation step involves two chemical reactions. The first reaction is initiation on homolytic fission in presence of high temperature to form free radicals

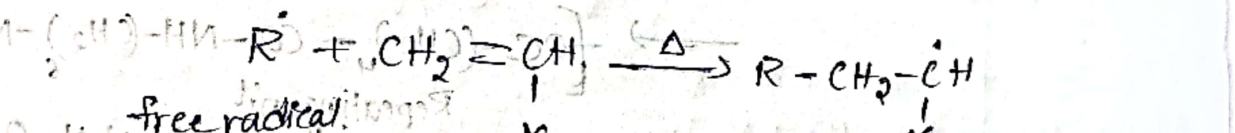






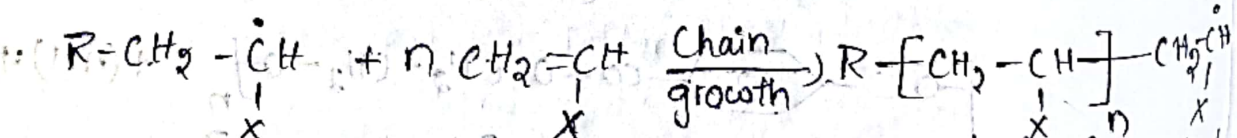
free radical.

In second reaction, free radical adds to vinyl monomer to form chain initiating free radicals.



free radical. Vinyl monomer Chain initiation free radical.

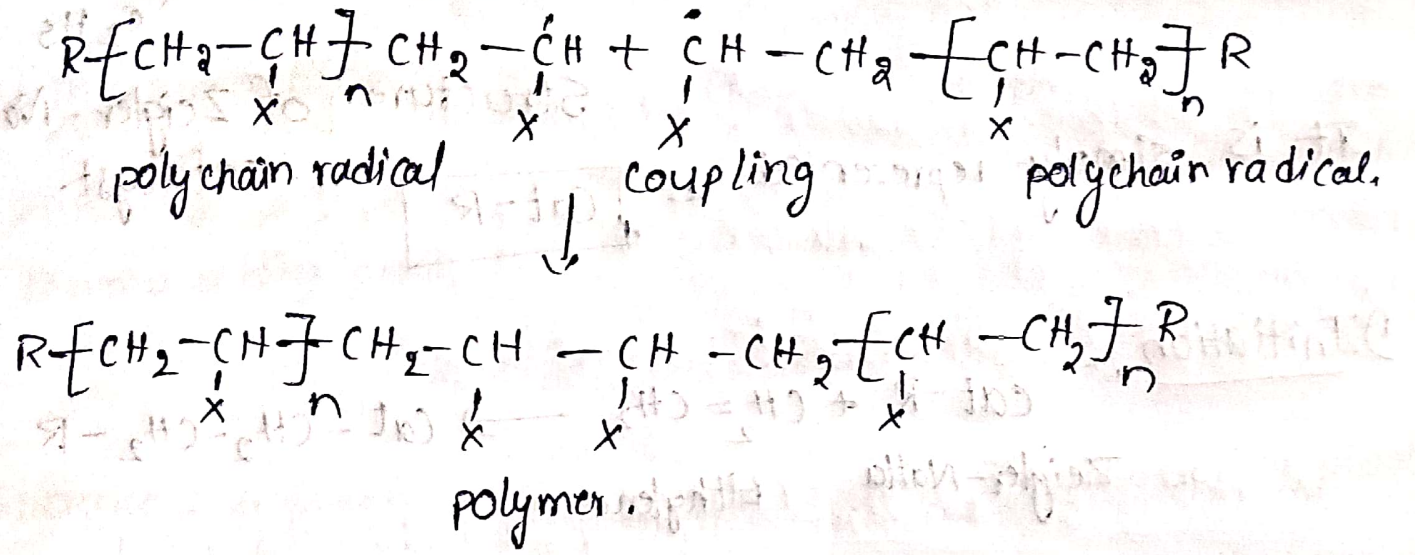
2) **Propagation**:- In propagation step, the chain initiating free radical on successive addition with member of vinyl monomers to form polychain free radical.



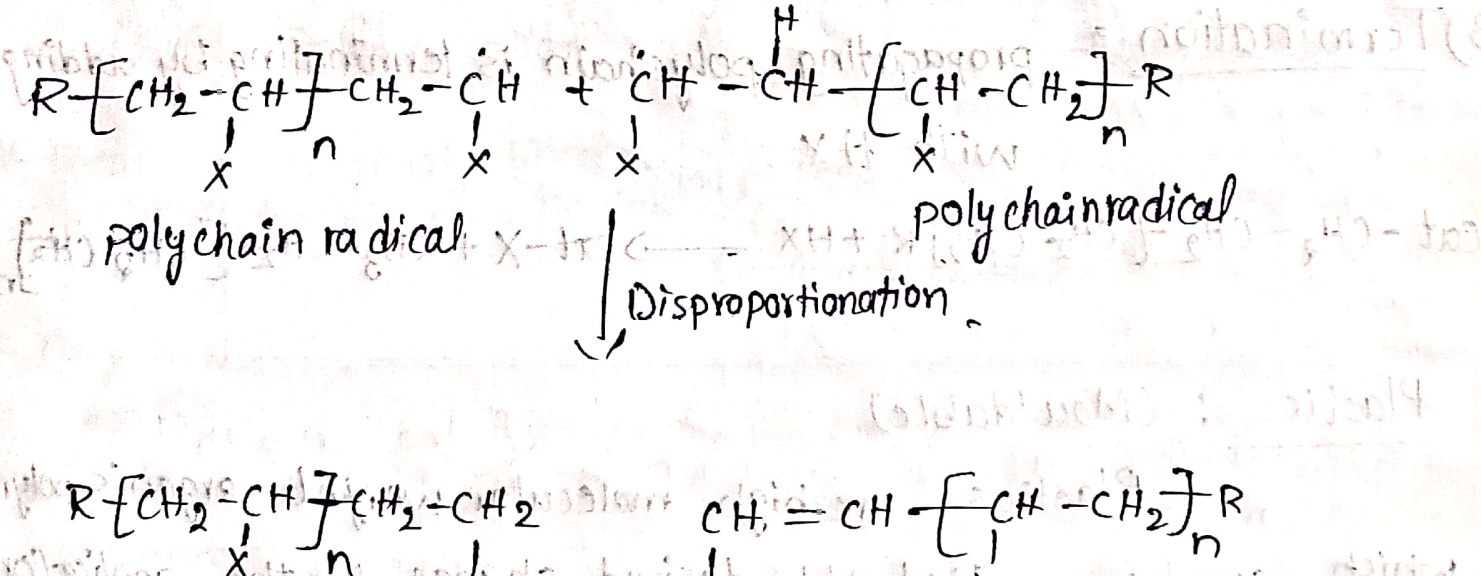
polymer free radical

3) **Terminating**:- In giving growing poly chain radical is terminated by conducting into dead polymer either by coupling or Disproportionation reaction.

1) Coupling :- In coupling polychain free radicals are joined end to end of the polychain radicals to form dead polymer.



2) Disproportionation :- In proportionation reaction, hydrogen atom of one polymer chain radical is transformed to another polymer chain radical to form one saturated Polymer and unsaturated polymer.



1. Thermo Plastics :- Plastics which soften on heating and harder on cooling. are called Thermo Plastics.

Ex:- Polythene, PVC, Teflon etc.

2. Thermo setting Plastics :- Plastics which become hard and rigid on heating and no change on cooling are called Thermo setting plastics.

Ex:- Bakelite, Urea-formaldehyde resin etc.

Differences between Thermoplastics and Thermosetting plastics.

Thermo Plastics	Thermo Setting Plastics.
* Plastics which soften on heating and harder on cooling are called Thermo plastics.	* Plastics which become hard & rigid on heating and no change on cooling are call Thermosetting.
* Most of them are formed by addition Polymerisation	* Most of these are formed by Condensation Polymerisation.
* These are long chain linear molecules with negligible cross-linking.	* These are three dimensional cross-linked molecules held by strong covalent bond.
* They are soft, weak and less brittle.	* They are hard, strong and more brittle.
* The linear chain are held by secondary forces.	* The polychains are held by covalent bonds.
* On reheating, they become soft and they can be reshaped and reuse	* On reheating they can not soften, they can not be reshaped and reused.
* They can be reclaimed from wastes	* they cannot be reclaimed from wastes.
* They are soluble in organic solvent	* They are insoluble in organic solvent
Ex:- PVC, Nylon 6, 6, poly ethylene etc	Ex:- Bakelite, Urea-formaldehyde resin etc.

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Fabrication of plastics :- The process of conversion of moulding constituents of plastics into articles is called Fabrication of Plastics.

Fabrication of plastic can be done by

① Compression moulding

② Injection moulding

(1) Compression moulding :-

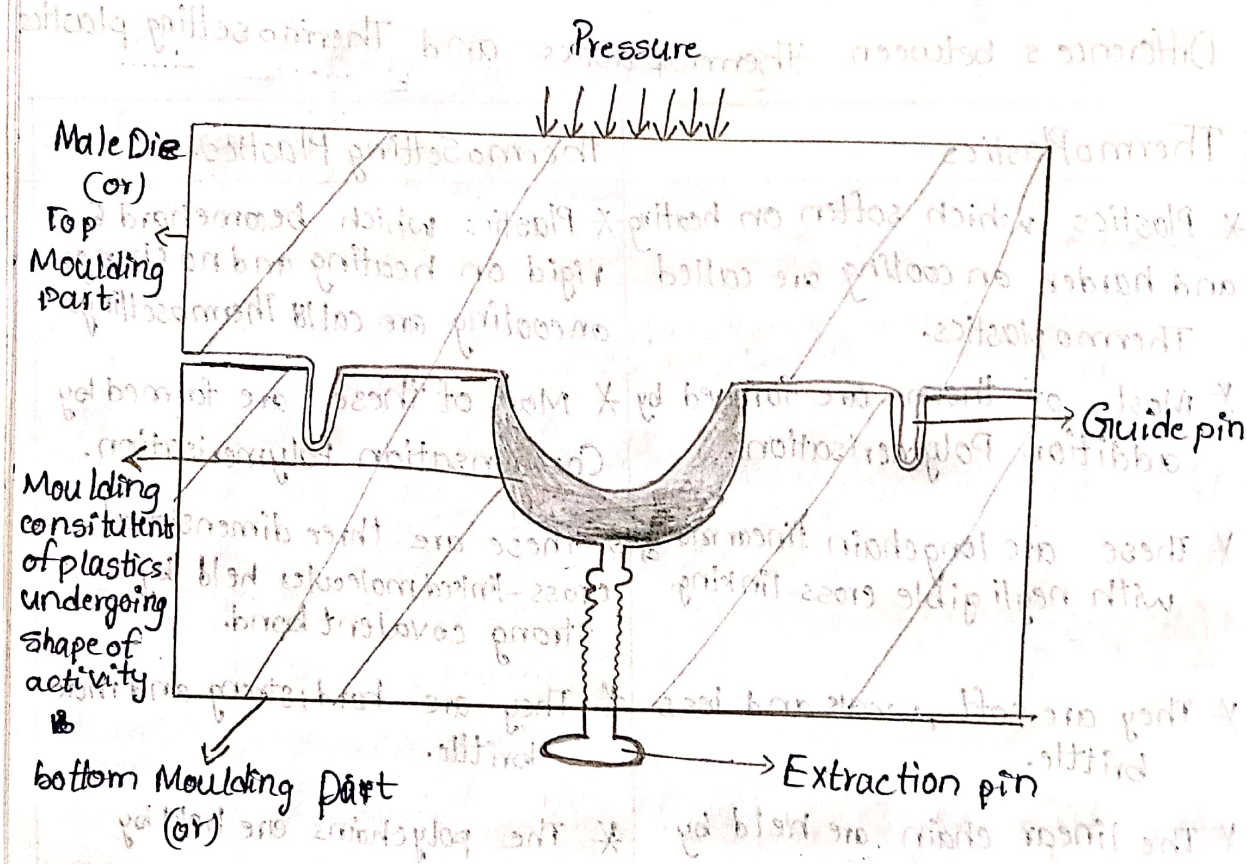
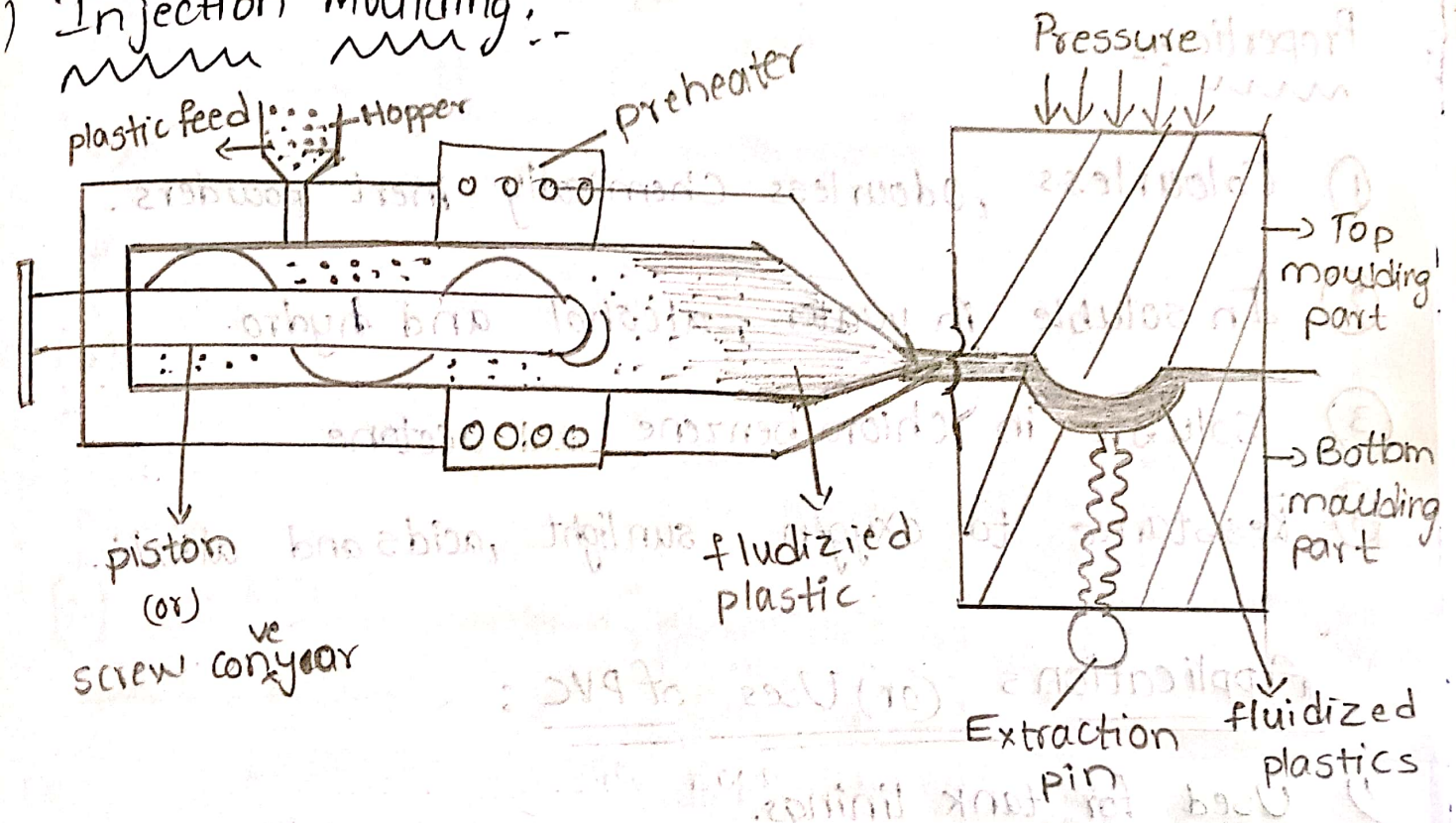


Fig: Compression Mailding

Compression Moulding is applicable for the both thermoplastics and thermosetting plastic. The plastic constituents are taking in the cavity of bottom moulding part and top moulding part is closed under pressure. In the case of thermo plastics curing is done by cooling. In the case of thermo setting plastics curing is done by heating. Now-a-day automobile compression Moulding machines are available for speed production of plastics.

(2) Injection Moulding :-



Injection Moulding is applicable for thermoplastics only. Moulding constiuents of plastics are fed through hopper into outer chamber, then heated, fludized plastic is formed. The fludized plastic is forced into mould by applying pressure through screw conveyor. The curving done by cooling, then particles is ejected from the mould

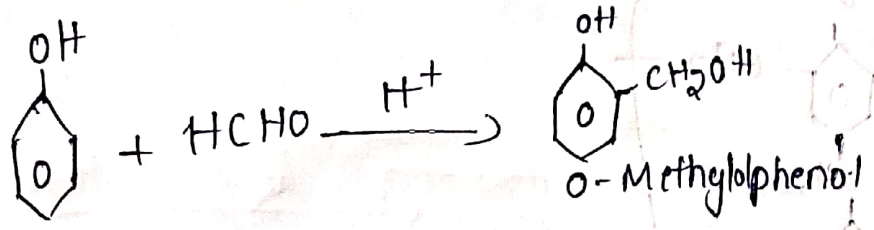
Bakelite :- Phenol formaldehyde resin is called Bakelite.

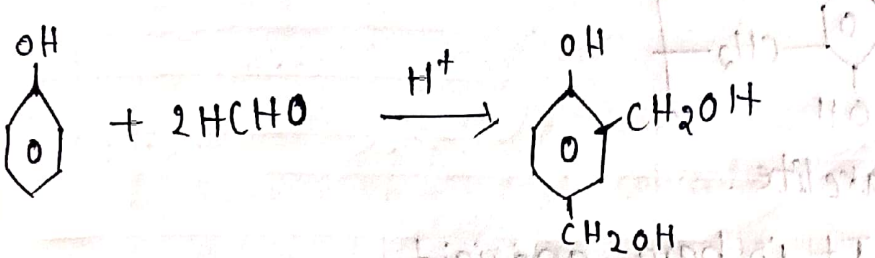
Preparation :

Poly condensation of phenol with formaldehyde gives Bakelite in 3 steps.

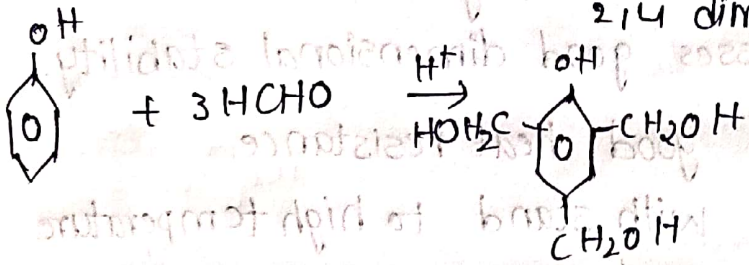
1st step :- When phenol is added to formaldehyde in presence of acid catalyst to form Methyl Phenol.

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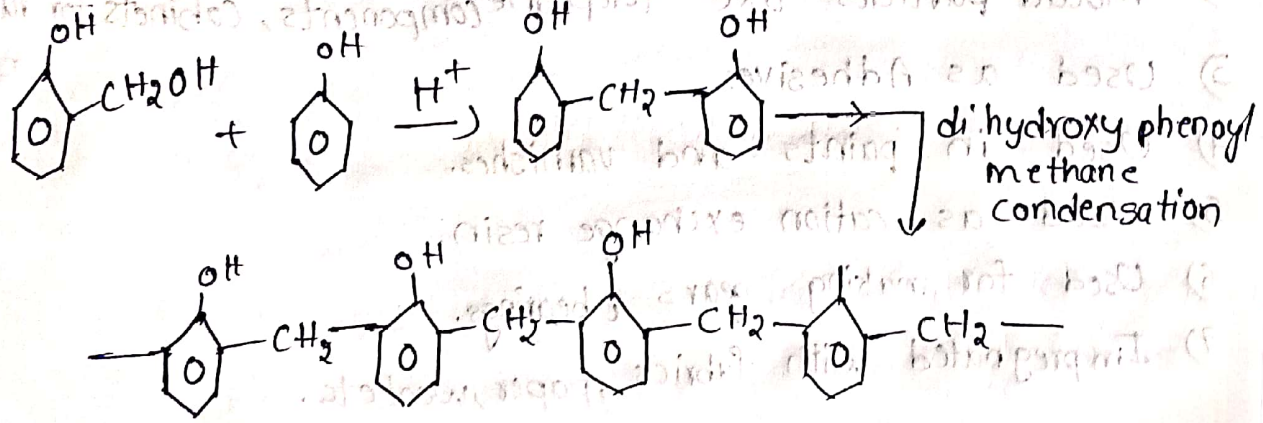
2,4 dimethyl phenol.



2,4,6 trimethyl phenol.

2nd step :- Novolac formation :

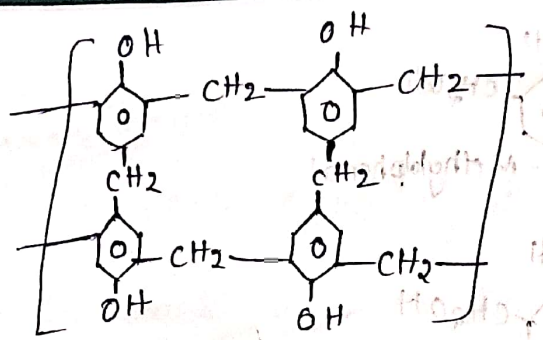
When phenol to formaldehyde ratio is greater than 1, in presence of acid catalyst to form dihydroxy diphenol methane, which further condenses with formaldehyde to form linear soluble polymer called "Novolac".



Novolac.

3rd step :- Bakelite :-

Novolac on curing with hexamethylene tetra amine to form three dimensional cross-linked polymer called "Bakelite".



Bakelite.

properties :- ① It is hard and rigid.
② It posses good dimensional stability.
③ It has good heat resistance.
④ It can with stand to high temperature
⑤ It has good Abrasive resistance.
⑥ It has good chemical resistance.
⑦ It has good electrical resistance & act as insulator.

Uses (or) Applications of Bakelite:-

① Electrical insulator parts like switches, sockets, plugs, switch boards, heater handles etc.
2) Mould particles like telephone components, cabinets for radio & TV.
3) Used as Adhesives.
4) Used in paints and varnishes.
5) Used as cation exchange resin.
6) Used for making gears & bearings.
7) Impregnated with fabrics, paper, wool etc.

Elastomer (or) Synthetic Rubber :-

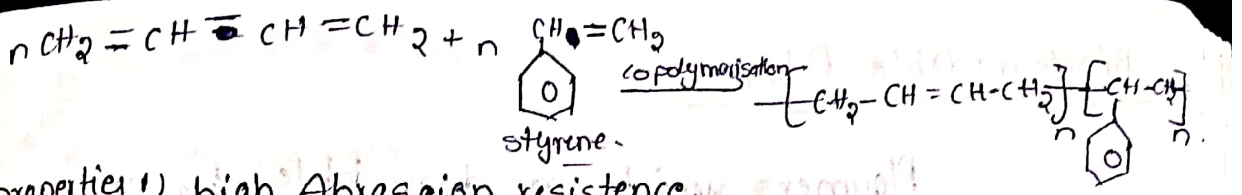
Elastomers are synthetic man made vulcanised rubber, which can be structured twice its length and it reaches original length when strech is released.

Ex:- ① BUNA-S ② BUNA-N

① **BUNA-S :-** BUNA-S is obtained by copolymerisation.

preparation :- 1,3 butadiene is reacted with styrene in presence of sodium catalyst.

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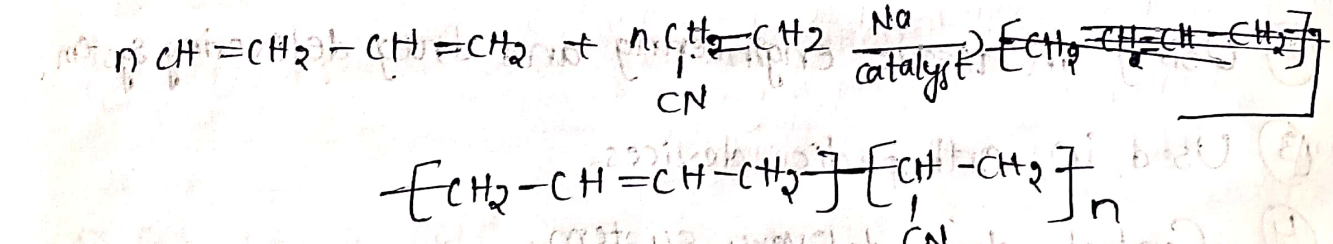


properties 1) high Abrasion resistence.
2) high load bearing capacity
3) good resistance
4) oxidises by atmospheric ozone (O_3).
5) swcele in oil and solvents.
6) Vulcanised by sulphur or sulphur monochloride.

Uses:- Used for manufacture of motor tyres, floor tiles, foot wear, shoes, adhesives, garkets, tank linings, carpet backings, insulating material etc.

2) BUNA-N (or) GIR-N (or) Nitrile rubber :-

It is obtained by copolymerisation of 1,3 butadiene and with acrylonitrle in presence of sodium catalyst.



Propatics:- (1) high Abrassion resistance
(2) high load bearing capacity.
(3) good resistance
(4) good resistance to acids, salts.
(5) resistance to oils and solvents.
(6) Vulcanized by sulphur or sulphur mono chloride.
(7) It is attacked alkali

Usey:- Used for manufacture of conveyor belts, hoses, ~~Aircraft~~ Aircraft components, Adhesives, garkets, insulating foams, tank linings, printing rollers etc.

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Biodegradable Polymers :

Ploymers which are degradable by enzymatic action of microorganism (or) bacteria are called bio-degradable polymers. Degradation of polymers by action of enzymes (or) Environmental factors like sunlight, temperature are called Bio-degradable polymers.

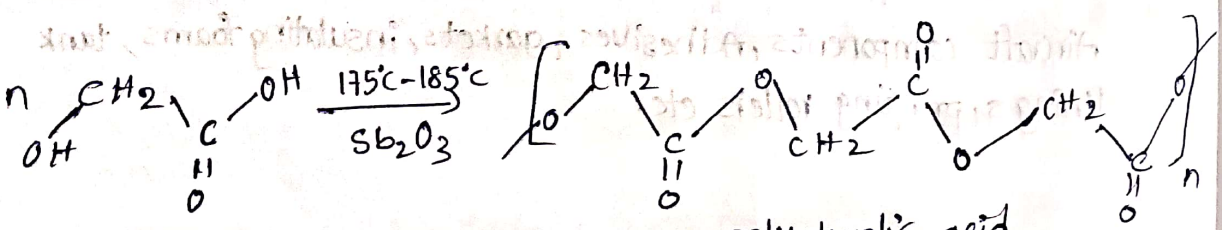
Ex:- Starch, cellulose, silk, wool, poly glycolic acid, poly lactic acids.

Applications of Biodegradable Polymers :

1. Extensively used in medical field
2. Using in tissue engineering and drug delivering system.
3. Used in orthopedic devices.
4. Control drug delivery system.
5. Dental implants, stents, screws, pins, mesh, Diapers, disposable garments, packing materials.

1) Polyglycolic acid (or) PGA : Polyglycolic acid is linear thermoplastic biodegradable polymer.

Preparation :- It is obtained by polycondensation of glycolic acid at 175°C – 185°C under 150mm pressure for 2 hours.



Glycolic acid

polyglycolic acid (PGA)

Properties:

(1) It has density 1.53 g/c.c

(2) M.P = 225°C – 230°C

(3) It decompes easily.

(4) due to presence of ester linkage which undergoes hydrolysis easily.

Uses of PGA :-

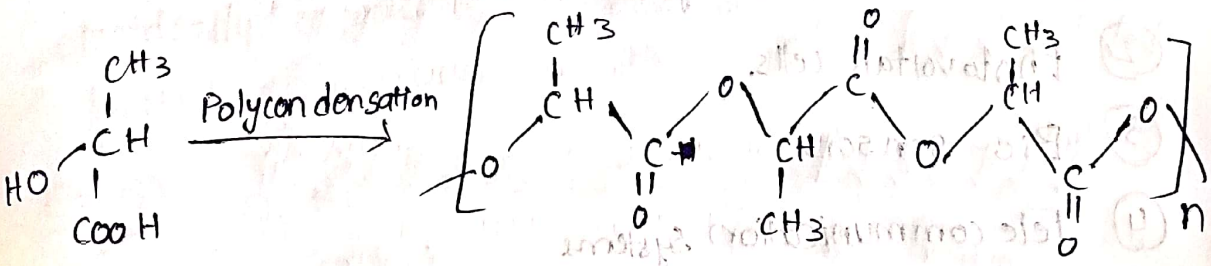
(1) Used for disposable sutures under trade name of 'Dexon'. Using in general surgery, Gynacology, Orthopaedic surgery, Urology etc.

(2) Used in food packing under trade name of 'Kure dux'.

(3) Used in oil and gas applications.

(2) Poly Lactic acid (PLA) : It is biodegradable thermoplastic ester.

Preparation :- It is obtained by polycondensation of Lactic acid at 175–178°C.



Lactic acid. poly Lactic acid

Properties :- (1) Its glass transistion temperature is 60°–66°C

(2) M.P is 178°–178°C

(3) Good bio compactiability

(4) Good processibility.

(5) High strength

(6) High Modules.

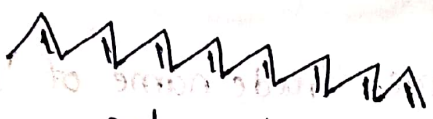
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Uses :- Used in Orthopaedic devices, control drug delivering systems, for making dental implants, anchors, screws, pins, Mesh, food packing materials, Diapers, Disposable garments etc.

Conducting Polymers :- Polymers which conduct electricity are called conducting polymers. Conducting polymers organic polymers having conjugated double bond sequences.

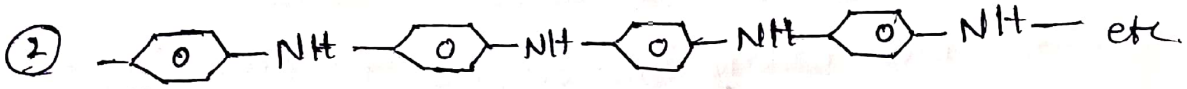
Ex:- Polyactylene, poly aniline, poly pyrrole, polythiophenone etc.

Ex:- ①



poly actylene

②



poly aniline

Applications of conducting polymers :-

Used in

① Used in Rechargable batteries.
② Photovoltaic cells.
③ Bio sensors.
④ Tele communication systems
⑤ Electronic devices like diodes, transistors.
⑥ Used in Light emitting diodes.
⑦ Electronic display boards
⑧ Micro chips.

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classification of Conducting Polymers:-

Conducting polymers

- Intrinsic Conducting polymers
- Extrinsic Conducting polymers.

Intrinsic Conducting polymers:- These are conjugated polymers which become electrically conductive due to presence of conjugated π-electrons on the polymer chain. These are 2 types.

1. Conjugated polymers with π electrons.
2. Doped Conjugated polymers.

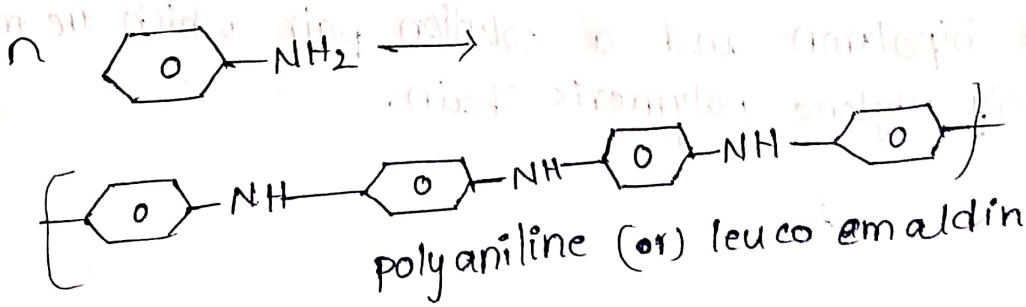
Conjugated π-electrons conducting polymer:-

These conducting polymers contain conjugated π electrons on polymer chain. The conjugated π-electrons are delocalised. In an electric field, the conjugated π-electrons transfered throughout the polymeric chain, which become electrically conductive. These conducting polymers have limited applications.

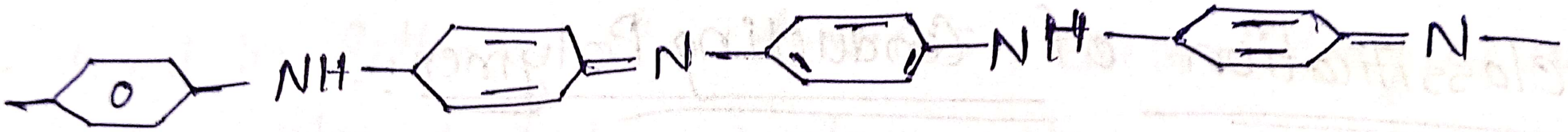
Ex (1) $n\ HC\equiv CH \longrightarrow$ 

Acetylene. polyacetylene

(2)



polyaniline (or) leuco emaldine.



Emaldine base.

ped Conducting Polymer :-

Doped Conducting polymers are obtained by
ucting polymer doping with either electron donor (or